

Management of Metastatic Clear Cell Renal Cell Carcinoma: ASCO Guideline Q&A

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INTRODUCTION

ASCO recently published a guideline on the management of metastatic clear cell renal cell carcinoma,¹ and three of those authors offer the following additional information in a Q&A format around the following topics: *What is the role of cytoreductive nephrectomy in metastatic clear cell renal cell carcinoma?*, *What are the preferred options for first-line systemic treatment of metastatic clear cell renal cell carcinoma?*, and finally *What is the optimal second or later line systemic treatment for metastatic clear cell renal cell carcinoma?* Fig 1 presents the summary of key guideline recommendations for the recommendations covered in this paper; the complete recommendations appear in the full guideline.¹

QUESTION: WHAT IS THE ROLE OF CYTOREDUCTIVE NEPHRECTOMY IN METASTATIC CLEAR CELL RENAL CELL CARCINOMA?

Cytoreductive nephrectomy (CN) remains a treatment option for selected patients with synchronous metastatic clear cell renal cell carcinoma (mRCC).^{2,3} It can also be a palliative option for hematuria or pain.⁴ CN should be considered for patients with one International mRCC Database Consortium (IMDC) risk factor (intermediate-risk disease).⁵ Although a definitive prognostication system for CN is lacking,⁶ optimal candidates for CN have the significant majority of their tumor burden within the kidney, have good Eastern Cooperative Oncology Group performance status, and do not have brain, bone, or liver metastases.⁷ IMDC poor-risk patients are likely to benefit more from immediate systemic therapy, although delayed CN can be considered with good response to systemic treatment. As with many complex operations, CN is best performed by high-volume surgeons at high-volume centers as part of multidisciplinary consultation.^{8,9} Enrollment in a CN clinical trial should be encouraged¹⁰ (Table 1).

QUESTION: WHAT ARE THE PREFERRED OPTIONS FOR FIRST-LINE SYSTEMIC TREATMENT OF mRCC?

Active surveillance may be considered for a subset of carefully selected patients with metastatic renal cell carcinoma who have a more indolent disease course.

Patients with lower IMDC risk score, limited number of metastatic sites, long intervals from nephrectomy to developing metastatic disease, and low disease volume might be better candidates for this approach. Metastasis-directed therapy such as surgery, radiation, or thermal ablation may be considered for select patients on surveillance.¹¹ Close monitoring for disease progression with regular serial imaging, including bone and CNS imaging, is essential to this approach. The benefit of preserving quality of life (QoL) and delaying or avoiding treatment-related side effects versus the potential for disease progression should be discussed with the patient to incorporate their preferences in the decision-making process.

For patients with IMDC intermediate- or poor-risk disease who are candidates for treatment with immune checkpoint inhibitors (ICIs), a combination regimen including an ICI should be used for the first-line systemic treatment. With five approved combination regimens (ipilimumab plus nivolumab, pembrolizumab plus axitinib, nivolumab plus cabozantinib, avelumab plus axitinib, and pembrolizumab plus lenvatinib), and in the absence of predictive biomarkers or prospective comparative data for these regimens, treatment selection should be based on side effects, comorbid conditions, provider experience, and out-of-pocket treatment costs. A detailed discussion with the patient and shared decision making is the recommended approach. Although QoL was evaluated in several of the studies, differences in QoL methodology and assessment limit the use of the information for decision making among the different treatment regimens.

Among the five approved combinations for the first-line treatment of mRCC, CheckMate-214 was designed with coprimary end points of overall response rate, progression-free survival, and overall survival (OS) among IMDC intermediate- and poor-risk patients. The secondary end points of this trial included overall response rate, progression-free survival, and OS in the intent-to-treat population that included the IMDC favorable-risk patients. The 42-month follow-up results of this study reported that the median OS was not reached in either arm (Nivo + Ipi v sunitinib) and the hazard ratio for death was 1.19 (95% CI, 0.77 to 1.85) among the exploratory efficacy population of

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Accepted on November 18, 2022 and published at ascopubs.org/journal-op on January 3, 2023; DOI <https://doi.org/10.1200/JOP.22.00660>

Recommendation	Type	Evidence Quality	Strength
2.1. Select patients (see Practical Information) with metastatic ccRCC may be offered cytoreductive nephrectomy. Practical Information: Select patients include those with optimally one IMDC risk factor who can have a significant majority of their tumor burden removed at the time of surgery.	EB	H	S
3.1. Select patients with metastatic ccRCC (see Practical Information) may be offered an initial active surveillance strategy. Practical Information: Select patients include those with IMDC favorable and intermediate risk, patients with limited or no symptoms related to disease, a favorable histologic profile, a long interval between nephrectomy and the development of metastasis, or with limited burden of metastatic disease.	EB	M	S
3.2. All patients with metastatic ccRCC who require systemic therapy in the first-line setting should undergo risk stratification into IMDC favorable (0), intermediate (1-2), and poor (3+) risk groups. Patients with intermediate- or poor-risk disease should be offered combination treatment with two ICIs (ie, ipilimumab and nivolumab) or an ICI in combination with a VEGFR TKI.	EB	H	S
3.3. Patients with favorable-risk disease who require systemic therapy may be offered an ICI in combination with a VEGFR TKI.	EB	H	S
3.4. Select patients with metastatic ccRCC receiving systemic therapy in the first-line setting including those with favorable-risk disease or with certain coexisting medical problems may be offered monotherapy with either a VEGFR TKI or an ICI.	EB	M	S
3.5. The use of HD-IL2 may be considered in the first-line systemic therapy setting for patients with metastatic ccRCC (see Practical Information). Attempts to develop criteria to predict those patients most likely to derive benefit from HD-IL2 have been unsuccessful. Practical Information: The significant toxicity of this regimen must be weighed in relation to the newer immunotherapy regimens that have largely replaced this treatment. The Expert Panel was not able to identify a patient population who should receive this treatment preferentially on the basis of available data. The Expert Panel did agree that HD-IL-2 should be administered in experienced high-volume centers, and that enrollment in clinical trials was preferred.	EB	M	W
4.1. Nivolumab or cabozantinib should be offered to patients who progressed on a VEGFR TKI alone.	EB	H	S
4.2. Patients progressing on combination immunotherapy (eg, nivolumab and ipilimumab) should be offered a VEGFR TKI.	CB	M	S
4.3. Patients who progress after initial therapy combining VEGFR TKI with an ICI may be offered an alternate VEGFR TKI as a single agent.	EB	H	S
4.4. For patients on immunotherapy who experience limited disease progression (eg, one site of progression), local therapy (radiation, thermal ablation, and excision) may be offered, and immunotherapy may be continued.	EB	M	W

FIG 1. Summary of key guideline recommendations. CB, consensus based; ccRCC, clear cell renal cell carcinoma; EB, evidence based; H, high; HD-IL, high-dose interleukin-2; ICI, immune checkpoint inhibitor; IMDC, International mRCC Database Consortium; M, moderate; S, strong; VEGFR TKI, vascular endothelial growth factor receptor tyrosine kinase inhibitor; W, weak.

favorable-risk patients. Therefore, the ipilimumab and nivolumab combination is currently approved for IMDC intermediate- and poor-risk patients and ICI + vascular endothelial growth factor receptor tyrosine kinase inhibitor (VEGFR TKI) combination regimens are recommended for patients with IMDC favorable-risk mRCC, in addition to the

intermediate- and poor-risk groups. Although combination therapy including ICIs is the preferred management strategy, use of VEGFR TKI monotherapy¹² or ICI monotherapy such as pembrolizumab¹³ may be considered in select patients. Risk group and/or coexisting medical issues may preclude use of ICIs or VEGFR TKIs (Fig 2).

TABLE 1. Ongoing Trials Looking at CN in the Immune Checkpoint Inhibitor Era

Study Name	Trial No.	Status	Primary End Point	Intervention
Deferred cytoreductive nephrectomy in synchronous metastatic renal cell carcinoma: The NORDIC-SUN-trial	NCT03977571	Recruiting	OS	Nivolumab, ipilimumab, CN
Comparing the outcome of immunotherapy-based drug combination therapy with or without surgery to remove the kidney in metastatic kidney cancer, the PROBE trial	NCT04510597	Recruiting	OS	CN, active comparator
CYTO-reductive surgery in kidney cancer plus immunotherapy and targeted kinase inhibition (CYTO-KIK)	NCT04322955	Recruiting	Complete response rate	Cabozantinib, nivolumab, CN
Nivolumab with or without bevacizumab or ipilimumab before surgery in treating patients with metastatic kidney cancer that can be removed by surgery	NCT02210117	Active, not recruiting	Adverse events	Bevacizumab, ipilimumab, nivolumab, metastasectomy, therapeutic conventional surgery, laboratory biomarker analysis, biopsy
Pembrolizumab with or without axitinib for treatment of locally advanced or metastatic clear cell kidney cancer in patients undergoing surgery	NCT04370509	Recruiting	Proportion of participants with > 2-fold increase in tumor-infiltrating immune cells	Axitinib, pembrolizumab, metastasectomy, CN

NOTE. Source: ClinicalTrials.gov. Accessed: September 14, 2022.

Abbreviations: CN, cytoreductive nephrectomy; OS, overall survival.

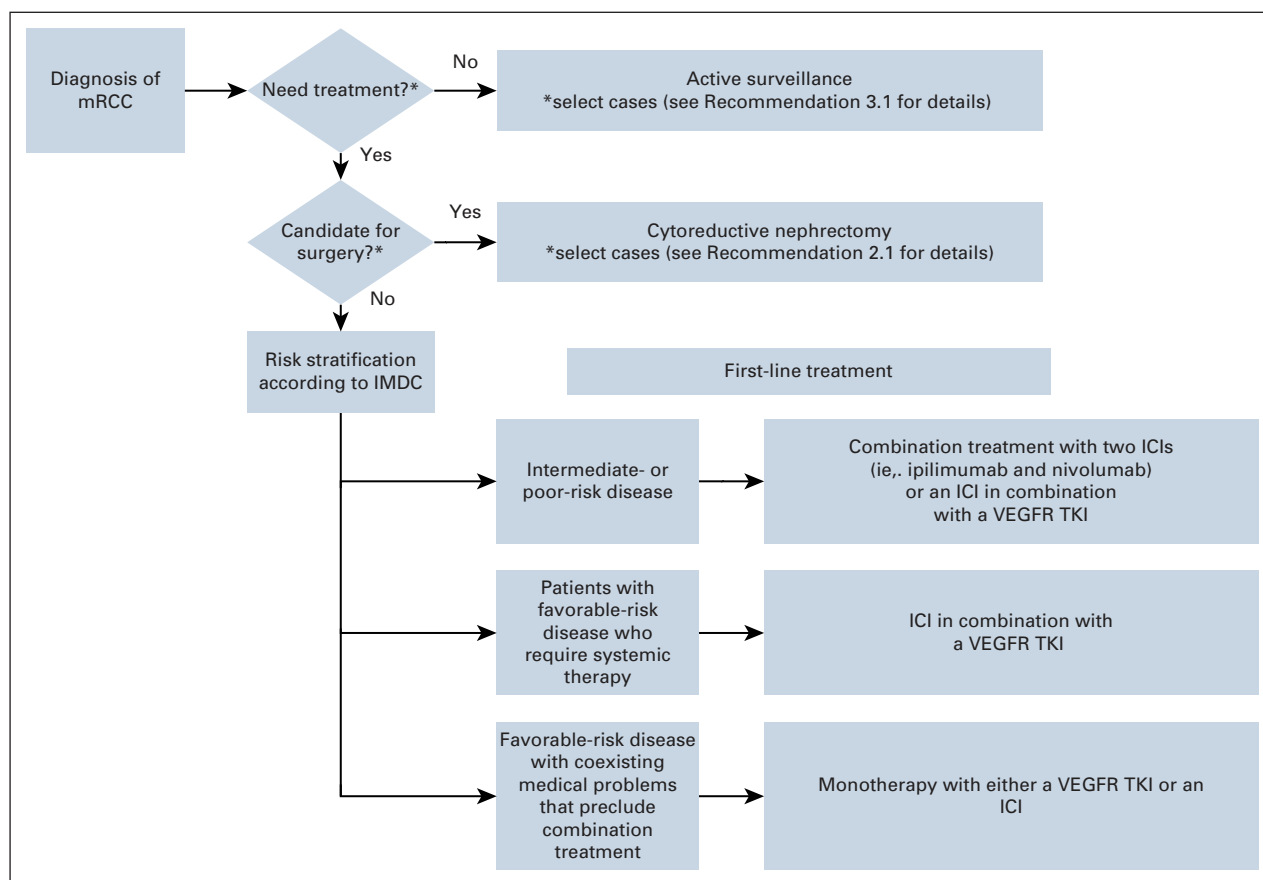


FIG 2. First-line treatment flowchart. ICI, immune checkpoint inhibitor; IMDC, International mRCC Database Consortium; mRCC, metastatic clear cell renal cell carcinoma; VEGFR TKI, vascular endothelial growth factor receptor tyrosine kinase inhibitor.

QUESTION: WHAT IS THE OPTIMAL SECOND OR LATER LINE SYSTEMIC TREATMENT FOR mRCC?

Most patients with metastatic renal cell carcinoma will experience disease progression on first-line therapy but will generally have an adequate performance status to consider second-line therapies or beyond. The choice of therapy will depend on which agents were used in first-line therapy, magnitude, and duration of response, toxicities, patient preferences, and comorbidities. Clinical trial participation is strongly suggested.¹⁰ Patients can be grouped into those who had TKI alone, immunotherapy alone (single agent or combination immunotherapy), or combination TKI with immunotherapy.

Considering that the combination of nivolumab and ipilimumab includes a drug from each of the main immunotherapy subclasses (anti-programmed cell death protein-1/programmed death ligand-1 and anti-cytotoxic T-cell lymphocyte 4), it is unlikely there will be a benefit to switching to a different immunotherapy drug of a similar class. Thus, switching to a VEGFR TKI (alone or in combination) would be preferred in this setting, and ongoing trials are testing this approach.

The increasing number of options for the treatment of metastatic renal cell cancer in the first-line setting makes the choice of treatment in the second-line setting dependent on what patients received in the first-line setting. What is equally important as patients move to second- and third-line treatment options is the integration of palliative care and supportive care measures (PSC).^{14,15} There are a variety of socioeconomic factors that affect the utilization and availability of these services that are still too frequently underutilized, and there is a need for the earlier initiation of PSC in a more equitable and defined fashion in patients with mRCC.¹⁶ One recent prostate cancer survey revealed that 86% of patients and caregivers reported this was an unmet need.¹⁶ However, there are an increasing number of tools to help better identify patients who are at high risk for needing PSC.¹⁷ Patients who do not respond well to first-line treatment have a significantly worse prognosis and represent a population of patients for whom PSC should be integrated as they move to second-line treatment options.¹⁸

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DISCLAIMER

This Q&A is derived from recommendations in Management of Metastatic Clear Cell Renal Cell Carcinoma: ASCO Guideline. This document is based on an ASCO Guideline and is not intended to substitute for the

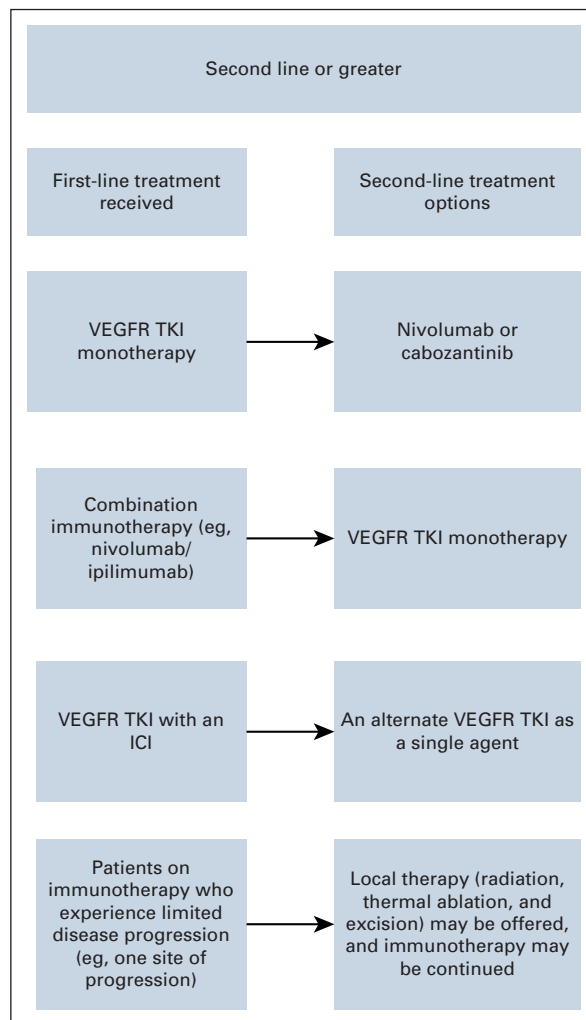


FIG 3. Second-line treatment flowchart. ICI, immune checkpoint inhibitor; VEGFR TKI, vascular endothelial growth factor receptor tyrosine kinase inhibitor.

Increasingly, there are geriatric assessment tools to assess frailty in elderly patients that can help both predict prognosis and toxicity risk,¹⁹ including an ASCO guideline on the assessment and management of vulnerabilities in older patients receiving chemotherapy²⁰ (Fig 3).

independent professional judgment of the treating physician. Practice guidelines do not account for individual variation among patients. This Q&A does not purport to suggest any particular course of medical treatment. Use of the guideline and this Q&A are voluntary. Please refer to the complete guideline to understand the full recommendations.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at DOI <https://doi.org/10.1200/OP.22.00660>.

AUTHOR CONTRIBUTIONS

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Administrative support: R. Bryan Rumble

Collection and assembly of data: All authors

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Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

ACKNOWLEDGMENT

Management of Metastatic Clear Cell Renal Cell Carcinoma: ASCO Guideline was developed and written by: W. Kimryn Rathmell, MD, PhD; R. Bryan Rumble, MSc; Peter J. Van Veldhuizen, MD; Hikmat Al-Ahmadie, MD; Hamid Emamekhoo, MD; Ralph J. Hauke, MD; Alexander V. Louie, MD, MSc, PhD; Matthew I. Milowsky, MD; Ana M. Molina, MD; Tracy L. Rose, MD, MPH; Shankar Siva, MD, PhD; Nicholas G. Zaorsky, MD, MS; Tian Zhang, MD; Rubina Qamar, MD; Terry M. Kungel, MBA; Bryan Lewis, JD; and Eric A. Singer, MD, MA, MS.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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Eric A. Singer

Consulting or Advisory Role: Merck, Johnson & Johnson/Janssen, Vyriad, Aura Biosciences

Research Funding: Astellas Medivation

No other potential conflicts of interest were reported.